MTH601 Assignment #01 2023

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Note: Check all calculations Before submitting if you found any mistake correct it yourself . for any query feel free to contact me.

Question #01:

If the production of an item is instantaneous with storage cost of Rs.5 per item per month. Other clerical and administrative costs associated to it are of Rs.30 per cycle. Find the best time for replacement provided that its demand is Rs.250 units of items per month.

Solution:

$Demand = D = Rs.250\ units\ \ per\ Month$

$Ordering\ \ \text{Cost} = C_2 = Rs.30\ \ per\ \ cycle$

$C_3 = Storage / Holding\ \ \text{Cost} = Rs.5\ \ per\ \ item / month$

$Time = ?$

Calculations:

$For\ \ finding\ \ time\ \ we\ \ have\ to\ \ calculate\ \ Q^*\left(EOQ\right)$

$$EOQ = Q^* = \sqrt{\frac{2C_2D}{C_3}} = \sqrt{\frac{2*30*250}{5}}$$

$$Q^* = \sqrt{\frac{1500}{5}} = \sqrt{3000} \approx 54.7722 \approx 55\ units$$

$Now\ \ for\ \ finding\ \ t\ \ we\ \ use:$

$$t = \frac{Q}{D} \approx \frac{55}{250} \approx 0.22$$

Question # 02:

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Demand for an item in a company is 5,000 units per year and company can produce at a rate of 1500 units per month. Cost of ordering is Rs 800 per setup and holding cost is 20 units per year, the shortage cost is 1000 per unit per year If optimum manufacturing quantity is 1400 then find the following

a. Time Between orders

b. Number of orders per year

Solution:

$Demand = D = 5000\ units / year$

$\Pr oduction\ rate = R = 1500\ units / month$

$Ordering\ \cos t = C_2 = Rs.800$

$Holding\ \cos t = C_3 = 20\ units / year$

$Shortage\ \cos t = C_4 = 1000\ units / year$

$Optimum\ manufacturing\ Quantity = Q^* = 1400$

Calculations:

- *Time between orders* :

$$t = \frac{Q}{D} = \frac{1400}{5000} = 0.28$$

- *Number of orders Per year* :

$$N = \frac{D}{Q} = \frac{5000}{1400} \approx 3.5714 \approx 4$$

- *Time between orders is* 0.28.
- *Number of orders per year is approximately* 4.

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